

Retrieval and Structuring Augmented Generation with LLMs for Web Applications (Lecture-Style)

Yizhu Jiao
yizhuj2@illinois.edu
University of Illinois Urbana-Champaign
Urbana, IL, USA

Siru Ouyang
siruo2@illinois.edu
University of Illinois Urbana-Champaign
Urbana, IL, USA

Ming Zhong
mingz5@illinois.edu
University of Illinois Urbana-Champaign
Urbana, IL, USA

Yunyi Zhang
yzhan238@illinois.edu
University of Illinois Urbana-Champaign
Urbana, IL, USA

Linyi Ding
linyid2@illinois.edu
University of Illinois Urbana-Champaign
Urbana, IL, USA

Sizhe Zhou
sizhez@illinois.edu
University of Illinois Urbana-Champaign
Urbana, IL, USA

Jiawei Han
hanj@illinois.edu
University of Illinois Urbana-Champaign
Urbana, IL, USA

Abstract

Although Large Language Models (LLMs) have revolutionized natural language processing, they face significant challenges in web applications, including hallucinations, outdated knowledge, and limited specialization in niche domains. To address these issues, this tutorial explores how integrating retrieval mechanisms and structured knowledge can enhance LLM performance for web use. By leveraging Retrieval-Augmented Generation (RAG), we can ground LLM outputs with relevant external data, mitigating limitations in applications like search engines, chatbots, and recommendation systems. We delve into text structuring techniques—such as taxonomy construction, multi-level text classification, and taxonomy-guided information retrieval—that improve the effectiveness of information retrieval processes. Furthermore, we examine how structure-guided augmented generation through information extraction and knowledge graph construction can reduce hallucinations and enhance factual accuracy. By bridging the gap between unstructured language models and structured knowledge, we aim to unlock new potentials for dynamic web content generation and personalized user experiences. Finally, we highlight future directions for seamlessly integrating retrieval and generation, enhancing personalization, and incorporating multimodal data to expand the capabilities of LLMs in web applications.

CCS Concepts

• **Computing methodologies** → **Natural language processing**.



This work is licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License.

WWW Companion '25, Sydney, NSW, Australia

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1331-6/2025/04
https://doi.org/10.1145/3701716.3715870

Keywords

Text Mining, Large Language Models, Retrieval Augmented Generation, Text Structuring

ACM Reference Format:
Yizhu Jiao, Siru Ouyang, Ming Zhong, Yunyi Zhang, Linyi Ding, Sizhe Zhou, and Jiawei Han. 2025. Retrieval and Structuring Augmented Generation with LLMs for Web Applications (Lecture-Style). In *Companion Proceedings of the ACM Web Conference 2025 (WWW Companion '25), April 28-May 2, 2025, Sydney, NSW, Australia.* ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3701716.3715870

Tutors and Past Tutorial Experiences

- **Yizhu Jiao**, Ph.D. candidate, Computer Science, UIUC. Her research focuses on knowledge structuring and grounding. She has received the Best Student Paper Runner-Up Award on ICDM 2020 and Best Reviewer Award on KDD 2023. She has delivered a tutorial in KDD'24.
- **Siru Ouyang**, Ph.D. candidate, Computer Science, UIUC. Her research focuses on mining and reasoning with structured knowledge from massive text data. She has received the Richard T. Cheng Fellowship (2022). She has many publications in ACL, ICML, EMNLP, and KDD. She has delivered a tutorial in KDD'24.
- **Ming Zhong**, Ph.D. candidate, Computer Science, UIUC. His research focuses on structuring explicit knowledge from massive corpora and manipulating implicit knowledge in foundation models. He has received the Amazon-Illinois Ph.D. Fellowship (2023). He published numerous research papers at ACL, EMNLP, NAACL, WWW, and ICLR. He has delivered a tutorial in KDD'24.
- **Yunyi Zhang**, Ph.D. candidate, Computer Science, UIUC. His research focuses on weakly supervised text mining, text classification, and taxonomy construction. He has numerous research publications at KDD, WWW, WSDM, ACL, and EMNLP and has delivered tutorials in EDBT'23, KDD'23, and KDD'24.

- **Linyi Ding**, M.S. student, Computer Science, UIUC. Her research focuses on weakly supervised knowledge graph construction and ontology construction from unstructured data. She has delivered a tutorial in KDD'24.
- **Sizhe Zhou**, M.S. student, Computer Science, UIUC. His research focuses on mining and understanding structured knowledge from massive text data. He has publications in EMNLP, ECML-PKDD, SDM, and EACL. He has delivered a tutorial in KDD'24.
- **Jiawei Han**, Michael Aiken Chair Professor, Computer Science, UIUC. His research areas encompass data mining, with over 1,000 research publications. He is the Fellow of ACM, Fellow of IEEE, and has received numerous prominent awards, including ACM SIGKDD Innovation Award (2004) and the IEEE Computer Society W. Wallace McDowell Award (2009). He has delivered 50+ conference tutorials or keynote speeches.

Topic and Relevance

The vast amount of web-based textual data contains invaluable and beneficial knowledge for various web-related tasks. However, effectively extracting and leveraging this knowledge from unstructured web text remains challenging. Our tutorial focuses on recent advancements in RAG and structured knowledge integration, which optimize LLM performance for web applications. By combining the strengths of LLMs with retrieval and strusturing methods, participants will unlock deeper insights from web-scale text data, which is crucial for tasks like dynamic content generation and web-based information retrieval. This tutorial is especially relevant for researchers and practitioners in web mining and natural language processing, equipping them with state-of-the-art tools to efficiently extract actionable knowledge from unstructured web content.

Target Audience

Researchers and practitioners in the fields of text mining, natural language processing, and information retrieval. While the audience with a good background in these areas would benefit most from this tutorial, we believe the material to be presented would give both general audience and newcomers an introductory pointer to the current work and important research topics in this field, and inspire them to learn more. Our tutorial is designed as self-contained, so only preliminary knowledge about basic concepts in data mining, text mining, machine learning, and their applications are needed.

Tutorial Schedule

Module I. Preliminary of Large Language Models (40 min)
Large Language Models (LLMs) leverage vast web data to generate and understand human language. The architectures of LLMs include encoder-only [10], encoder-decoder [33], and decoder-only models [24]. Pre-trained language models are adapted using supervised fine-tuning, instruction tuning [28], and Reinforcement Learning from Human Feedback (RLHF) [35]. Additionally, prompting strategies such as In-Context Learning (ICL) [3] and Chain-of-Thought (CoT) [39] enhance their performance. Despite their advancements, LLMs face challenges in web applications: *(i) Hallucination:* Generating responses that sound plausible but are factually inaccurate or misleading [15]. *(ii) Untimely knowledge update:* Lack of awareness of recent events due to training on static data [4]. *(iii) Insufficient coverage of specialized domains:* Limited performance in niche areas lacking sufficient training data [20].
Module II. Retrieval-Augmented Generation (30 min)
Large amounts of web content would be easier and more effective to digest with a *retrieval* step. RAG was initially developed to enhance PLMs by refining pre-training techniques with additional knowledge [2, 14]. The subsequent arrival of LLMs marked its superior ability of in-context learning (ICL), where RAG is leveraged to provide better information for LLMs during the inference stage [40, 46]. More recently, RAG has been integrated with LLM fine-tuning setups [26, 45]. By referencing external knowledge, RAG is usually integrated with language models for web-related real-world applications [32], specifically with domain-specific and knowledge-intensive tasks. We introduce the recent use cases of RAG techniques in web applications, including web search [41], chatbots [31], and recommendations [11]. Despite the wide usage of web applications, adopting RAG in web scenarios still poses challenges. We discuss the following three key challenges: spanning insufficient relevance of retrieved data [34], redundant or overlapping information [38], and inconsistent or fictitious contexts [43].
Module III. Text Structuring for Effective Retrieval (40 min)
(1) **Taxonomy Construction and Enrichment** Taxonomy construction and enrichment aim to create a hierarchical structure of concepts and enrich it with rich textual features to help interpret the structure. While existing taxonomies are mostly human curated and hard to adapt, automated taxonomy construction create such a structure using a set of seeds [19]. Taxonomy enrichment aims to discover a set of topical keywords that distinctively identify each node. GTM [9] proposes a seed-guided topic-noise model for short texts. SeedTopicMine [49] combines multiple contexts including PLM representations and topic-indicative text snippets.
(2) **Multi-Level Text Classification** Classification of web text corpus is a crucial task for web content analysis and semantic indexing. We introduce the weakly-supervised text classification setting, where only the class label names or a set of keywords are needed for classifier training. We will first introduce weakly-supervised flat text classification [47]. Then, we also introduce the hierarchical setting where the label space is a taxonomy and each document can have multiple labels at different levels [48]
(3) **Taxonomy-Guided Information Retrieval** Recent advancements on information retrieval include novel training [25], encoding strategies [42], and broader training methods [5]. For better adaptation on specialized corpora, we focuses on leverage the thematic structures from the constructed taxonomy as guidance. With the ability to construct or expand a topic taxonomy for specific themes and label all the documents with minimal human efforts, the fine-grained topic structure can be utilized to optimize the search space [22] and enhance the query-document matching [21].
Module IV. Structure-Guided Augmented Generation (40 min)
(1) **Information Extraction.** Named entity recognition (NER) locates and classifies named entities in unstructured text into predefined categories as a first step in information extraction. Recent studies [37] explored NER with the help of LLMs. Entity typing, on the other hand, focuses on identifying the type of a given entity in its context. Fine-grained entity typing [8] can be performed with the help of an entity taxonomy [29], in specific domains [50]. Relation extraction (RE) is the task aiming at identifying semantic

relationships between entities [7]. Combined with NER and entity typing, RE helps extract and organize structured knowledge from unstructured sources (e.g., text). In the era of LLMs, there are considerable advancements on synthesis [36] or relation understanding enhancement [27]. We will focus on recent approaches that leverage LLMs to minimize human supervision for efficient and versatile relation extraction [51], and balance the generative and ranking capabilities of LLMs for open relation extraction [12].

(2) **Knowledge Graph Construction.** Knowledge graphs (KG) represent the relations among various entities in a graph-like format, where nodes denote entities, and edges denote relations. KGs fall mainly into three categories, including general-purpose KGs, domain-specific KGs [1], and theme-specific KGs [13] for fine-gained purposes. KG construction is a systematic process that combines various information extraction methods. Recently, LLMs have emerged as powerful generative tools in the KG construction pipeline, especially in complex contexts. LLMs can be leveraged in multiple ways to facilitate KG construction including working as end-to-end agents in a zero-shot or few-shot manner [44], being agents for sub-tasks in the pipeline [6], and indirectly improving the tasks by discovering more supportive information [16].

(3) **Integrated Retrieval and Structuring to Augment Generation.** While LLMs may hallucinate in specialized and up-to-date facts, the KGs can provide factual knowledge and reasoning capabilities. Incorporating structured knowledge into LLMs may be one of the effective solutions to alleviating hallucination. The tutorial will introduce some effective works that improve LLMs by retrieval of structured knowledge on several tasks (e.g. entity typing, text summarization [18], reasoning [17]). These methods select supportive evidence from the structured knowledge to improve the knowledge-grounding ability of LLMs. The KGs also can be applied to the LLMs' pertaining phase [23] and LLMs' interpretability [30].

Module V. Web Applications and Future Directions (20 min)

(1) **Applications in Web Systems** (i) Dynamic Web Content Generation. Using structured augmented generation to dynamically produce timely and contextually relevant content, adapting in real-time to user behavior and external data sources. (ii) Personalized Chat Bots. Developing advanced chat bots that leverage structured KGs and retrieval-augmented techniques for personalized and intuitive interactions. These bots understand user preferences and history to provide tailored recommendations, support, and services.

(2) **Future Directions** (i) Improved Integration of Retrieval and Generation. Developing seamless knowledge fusion techniques will naturally incorporate retrieved information into generated text, enhancing accuracy and reducing hallucinations. (ii) Personalization and User Adaptation. Adaptive systems that learn from user interactions can refine RSAG performance, making generated content better aligned with user needs. (iii) Multimodal Retrieval and Generation. Integrating text, images, audio, and video enables models to process and generate multimodal content, broadening applications to areas like video summarization and multimedia creation.

Previous Editions

One related tutorial with the same authors is titled as **Automated Mining of Structured Knowledge from Text in the Era of Large Language Models** [1] at **KDD'24**. The difference between these two versions is the expanded scope to include retrieval-augmented techniques and structure-guided generation for web applications. While the KDD'24 tutorial focused on mining structured knowledge using LLMs with weak supervision, this updated version emphasizes integrating LLMs with retrieval mechanisms like RAG and taxonomy-guided retrieval. It addresses additional challenges such as mitigating hallucinations, incorporating up-to-date knowledge, and enhancing domain-specific performance.

Acknowledgement

Research was supported in part by US DARPA INCAS Program No. HR0011-21-C0165 and BRIES Program No. HR0011-24-3-0325, National Science Foundation IIS-19-56151, the Molecule Maker Lab Institute: An AI Research Institutes program supported by NSF under Award No. 2019897, and the Institute for Geospatial Understanding through an Integrative Discovery Environment (I-GUIDE) by NSF under Award No. 2118329.

References

[1] Bilal Abu-Salih. 2021. Domain-specific knowledge graphs: A survey. *Journal of Network and Computer Applications* 185 (2021), 103076.

[2] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George Bm Van Den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, et al. 2022. Improving language models by retrieving from trillions of tokens. In *International conference on machine learning*. PMLR, 2206–2240.

[3] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language Models are Few-Shot Learners. In *NeurIPS*.

[4] Nicola De Cao, Wilker Aziz, and Ivan Titov. 2021. Editing Factual Knowledge in Language Models. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, EMNLP 2021, Virtual Event / Punta Cana, Dominican Republic, 7-11 November, 2021*, Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih (Eds.). Association for Computational Linguistics, 6491–6506. doi:10.18653/V1/2021.EMNLP-MAIN.522

[5] Aditi Chaudhary, Karthik Raman, Krishna Srinivasan, Kazuma Hashimoto, Mike Bendersky, and Marc Najork. 2023. Exploring the viability of synthetic query generation for relevance prediction. *arXiv preprint arXiv:2305.11944* (2023).

[6] Jiao Chen, Luyi Ma, Xiaohan Li, Nikhil Thakurdesai, Jianpeng Xu, Jason HD Cho, Kaushiki Nag, Evren Korpeoglu, Sushant Kumar, and Kannan Achan. 2023. Knowledge Graph Completion Models are Few-shot Learners: An Empirical Study of Relation Labeling in E-commerce with LLMs. *arXiv preprint arXiv:2305.09858* (2023).

[7] Yew Ken Chia, Lidong Bing, Soujanya Poria, and Luo Si. 2022. RelationPrompt: Leveraging prompts to generate synthetic data for zero-shot relation triplet extraction. *arXiv preprint arXiv:2203.09101* (2022).

[8] Eunsol Choi, Omer Levy, Yejin Choi, and Luke Zettlemoyer. 2018. Ultra-Fine Entity Typing. In *ACL*.

[9] Robert Churchill, Lisa Singh, Rebecca Ryan, and Pamela Davis-Kean. 2022. A Guided Topic-Noise Model for Short Texts. In *WWW*.

[10] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *NAACL-HLT*. 4171–4186.

[11] Dario Di Palma. 2023. Retrieval-augmented recommender system: Enhancing recommender systems with large language models. In *Proceedings of the 17th ACM Conference on Recommender Systems*. 1369–1373.

[12] Linyi Ding, Jinfeng Xiao, Sizhe Zhou, Chaoqi Yang, and Jiawei Han. 2024. Topic-Oriented Open Relation Extraction with A Priori Seed Generation. In *EMNLP*.

[1] https://yzhan238.github.io/kdd24-tutorial/

[13] Linyi Ding, Sizhe Zhou, Jinfeng Xiao, and Jiawei Han. 2024. Automated Construction of Theme-specific Knowledge Graphs. arXiv:2404.19146 [cs.AI] https://arxiv.org/abs/2404.19146

[14] Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. 2020. REALM: retrieval-augmented language model pre-training. In *Proceedings of the 37th International Conference on Machine Learning (ICML'20)*. JMLR.org, Article 368, 10 pages.

[15] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. 2023. A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. *CoRR* abs/2311.05232 (2023). doi:10.48550/ARXIV.2311.05232 arXiv:2311.05232

[16] Pengcheng Jiang, Shivam Agarwal, Bowen Jin, Xuan Wang, Jimeng Sun, and Jiawei Han. 2023. Text-augmented open knowledge graph completion via pre-trained language models. *arXiv preprint arXiv:2305.15597* (2023).

[17] Pengcheng Jiang, Cao Xiao, Minhao Jiang, Parminder Bhatia, Taha Kass-Hout, Jimeng Sun, and Jiawei Han. 2024. Reasoning-Enhanced Healthcare Predictions with Knowledge Graph Community Retrieval. *arXiv preprint arXiv:2410.04585* (2024).

[18] Pengcheng Jiang, Cao Xiao, Zifeng Wang, Parminder Bhatia, Jimeng Sun, and Jiawei Han. 2024. TriSum: Learning Summarization Ability from Large Language Models with Structured Rationale. *arXiv preprint arXiv:2403.10351* (2024).

[19] Song Jiang, Qiyue Yao, Qifan Wang, and Yizhou Sun. 2023. A Single Vector Is Not Enough: Taxonomy Expansion via Box Embeddings. In *WWW*. 2467–2476.

[20] Nikhil Kandpal, Haikang Deng, Adam Roberts, Eric Wallace, and Colin Raffel. 2023. Large Language Models Struggle to Learn Long-Tail Knowledge. In *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA (Proceedings of Machine Learning Research, Vol. 202)*, Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (Eds.). PMLR, 15696–15707. https://proceedings.mlr.press/v202/kandpal23a.html

[21] SeongKu Kang, Shivam Agarwal, Bowen Jin, Dongha Lee, Hwanjo Yu, and Jiawei Han. 2024. Improving Retrieval in Theme-specific Applications using a Corpus Topical Taxonomy. In *Proceedings of the ACM on Web Conference 2024*. 1497–1508.

[22] Yueqing Liang, Liangwei Yang, Chen Wang, Xiongxiao Xu, Philip S Yu, and Kai Shu. 2024. Taxonomy-Guided Zero-Shot Recommendations with LLMs. *arXiv preprint arXiv:2406.14043* (2024).

[23] Weijie Liu, Peng Zhou, Zhe Zhao, Zhiruo Wang, Qi Ju, Haotang Deng, and Ping Wang. 2020. K-bert: Enabling language representation with knowledge graph. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 34. 2901–2908.

[24] Llama Team. 2024. The Llama 3 Herd of Models. *CoRR* abs/2407.21783 (2024). doi:10.48550/ARXIV.2407.21783 arXiv:2407.21783

[25] Yi Luan, Jacob Eisenstein, Kristina Toutanova, and Michael Collins. 2021. Sparse, dense, and attentional representations for text retrieval. *Transactions of the Association for Computational Linguistics* 9 (2021), 329–345.

[26] Xueguang Ma, Liang Wang, Nan Yang, Furu Wei, and Jimmy Lin. 2024. Fine-tuning llama for multi-stage text retrieval. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2421–2425.

[27] Yubo Ma, Yixin Cao, YongChing Hong, and Aixin Sun. 2023. Large language model is not a good few-shot information extractor, but a good reranker for hard samples! *arXiv preprint arXiv:2303.08559* (2023).

[28] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F. Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *NeurIPS*.

[29] Siru Ouyang, Jiaxin Huang, Pranav Pillai, Yunyi Zhang, Yu Zhang, and Jiawei Han. 2024. Ontology Enrichment for Effective Fine-grained Entity Typing. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, KDD 2024, Barcelona, Spain, August 25-29, 2024*, Ricardo Baeza-Yates and Francesco Bonchi (Eds.). ACM, 2318–2327. doi:10.1145/3637528.3671857

[30] Fabio Petroni, Tim Rocktäschel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, Alexander H Miller, and Sebastian Riedel. 2019. Language models as knowledge bases? *arXiv preprint arXiv:1909.01066* (2019).

[31] Zackary Rackauckas. 2024. Rag-fusion: a new take on retrieval-augmented generation. *arXiv preprint arXiv:2402.03367* (2024).

[32] Irina Radeva, Ivan Popchev, Lyubka Doukovska, and Miroslava Dimitrova. 2024. Web Application for Retrieval-Augmented Generation: Implementation and Testing. *Electronics* 13, 7 (2024), 1361.

[33] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. *J. Mach. Learn. Res.* 21 (2020), 140:1–140:67.

[34] Alireza Salemi and Hamed Zamani. 2024. Evaluating retrieval quality in retrieval-augmented generation. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2395–2400.

[35] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal Policy Optimization Algorithms. *arXiv preprint arXiv:1707.06347* (2017).

[36] Qi Sun, Kun Huang, Xiaocui Yang, Rong Tong, Kun Zhang, and Soujanya Poria. 2024. Consistency Guided Knowledge Retrieval and Denoising in LLMs for Zero-shot Document-level Relation Triplet Extraction. *arXiv preprint arXiv:2401.13598* (2024).

[37] Shuhe Wang, Xiaofei Sun, Xiaoya Li, Rongbin Ouyang, Fei Wu, Tianwei Zhang, Jiwei Li, and Guoyin Wang. 2023. GPT-NER: Named Entity Recognition via Large Language Models. *CoRR* abs/2304.10428 (2023). doi:10.48550/ARXIV.2304.10428 arXiv:2304.10428

[38] Xiaohua Wang, Zhenghua Wang, Xuan Gao, Feiran Zhang, Yixin Wu, Zhibo Xu, Tianyuan Shi, Zhengyuan Wang, Shizheng Li, Qi Qian, Ruicheng Yin, Changze Lv, Xiaoqing Zheng, and Xuanjing Huang. 2024. Searching for Best Practices in Retrieval-Augmented Generation. *CoRR* abs/2407.01219 (2024). doi:10.48550/ARXIV.2407.01219 arXiv:2407.01219

[39] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In *NeurIPS*.

[40] Shi Weijia, Min Sewon, Yasunaga Michihiro, Seo Minjoon, James Rich, Lewis Mike, and Yih Wen-tau. 2023. REPLUG: Retrieval-augmented black-box language models. *arXiv preprint ArXiv:2301.12652* (2023).

[41] Weijian Xie, Xuefeng Liang, Yuhui Liu, Kaihua Ni, Hong Cheng, and Zetian Hu. 2024. WeKnow-RAG: An Adaptive Approach for Retrieval-Augmented Generation Integrating Web Search and Knowledge Graphs. *arXiv preprint arXiv:2408.07611* (2024).

[42] Lee Xiong, Chenyan Xiong, Ye Li, Kwok-Fung Tang, Jialin Liu, Paul Bennett, Junaid Ahmed, and Arnold Overwijk. 2020. Approximate nearest neighbor negative contrastive learning for dense text retrieval. *arXiv preprint arXiv:2007.00808* (2020).

[43] Yunqi Xu, Tianchi Cai, Jiyan Jiang, and Xierui Song. 2024. Face4RAG: Factual Consistency Evaluation for Retrieval Augmented Generation in Chinese. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 6083–6094.

[44] Bowen Zhang and Harold Soh. 2024. Extract, Define, Canonicalize: An LLM-based Framework for Knowledge Graph Construction. *arXiv preprint arXiv:2404.03868* (2024).

[45] Peitian Zhang, Shitao Xiao, Zheng Liu, Zhicheng Dou, and Jian-Yun Nie. 2023. Retrieve anything to augment large language models. *arXiv preprint arXiv:2310.07554* (2023).

[46] Tianjun Zhang, Shishir G. Patil, Naman Jain, Sheng Shen, Matei Zaharia, Ion Stoica, and Joseph E. Gonzalez. 2024. RAFT: Adapting Language Model to Domain Specific RAG. *CoRR* abs/2403.10131 (2024). doi:10.48550/ARXIV.2403.10131 arXiv:2403.10131

[47] Yunyi Zhang, Minhao Jiang, Yu Meng, Yu Zhang, and Jiawei Han. 2023. PIEClass: Weakly-Supervised Text Classification with Prompting and Noise-Robust Iterative Ensemble Training. In *EMNLP*. 12655–12670.

[48] Yunyi Zhang, Ruozhen Yang, Xueqiang Xu, Rui Li, Jinfeng Xiao, Jiaming Shen, and Jiawei Han. 2024. TELEClass: Taxonomy Enrichment and LLM-Enhanced Hierarchical Text Classification with Minimal Supervision. *arXiv preprint arXiv:2403.00165* (2024).

[49] Yu Zhang, Yunyi Zhang, Martin Michalski, Yucheng Jiang, Yu Meng, and Jiawei Han. 2023. Effective Seed-Guided Topic Discovery by Integrating Multiple Types of Contexts. In *WSDM*.

[50] Yu Zhang, Yunyi Zhang, Yanzhen Shen, Yu Deng, Lucian Popa, Larisa Shwartz, ChengXiang Zhai, and Jiawei Han. 2024. Seed-Guided Fine-Grained Entity Typing in Science and Engineering Domains. In *Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence, IAAI 2024, Fourteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2014, February 20-27, 2024, Vancouver, Canada*, Michael J. Wooldridge, Jennifer G. Dy, and Sriraam Natarajan (Eds.). AAAI Press, 19606–19614. doi:10.1609/AAAI.V38I17.29933

[51] Sizhe Zhou, Yu Meng, Bowen Jin, and Jiawei Han. 2024. Grasping the Essentials: Tailoring Large Language Models for Zero-Shot Relation Extraction. In *EMNLP*.